Parsa Esmaeilkhani

445.207.1071 | parsa.esmaeilkhani@temple.edu | LinkedIn | Google Scholar
Philadelphia, PA

SUMMARY

PhD candidate in Computer Science, expected to graduate in Summer 2027, specializing in computer vision, vision foundation models, vision-language models, and visual grounding. Experience in deep learning for computational pathology at Mayo Clinic and audio processing at Fraunhofer IIS. Seeking internships in computer vision and multimodal AI for Summer 2027

EDUCATION

PhD Candidate in Computer Science (GPA: 3.88/4.0) January 2023 – July 2027 (Expected)
Artificial Intelligence Division - Temple University Philadelphia, PA

Dissertation: *Grounding Visual Information in Vision-Language Models.*

Research Interests: Vision-Language Models, Visual Grounding, Multimodal Representation Learning, Visual Explainability.

Bachelor of Science in Mathematics and Applications September 2016 – June 2021
University of Tehran Tehran, Iran

INDUSTRY EXPERIENCE

AI & Data Science Research Intern January 2026 – May 2026
Mayo Clinic Rochester, MN, USA

- Conducted computational pathology research using private Mayo Clinic breast pathology datasets and a curated WHO reference atlas for fine-grained retrieval and classification.
- Pretrained and fine-tuned vision foundation models using self-supervised and supervised learning on institution-specific whole-slide image datasets.
- Developed and evaluated pathology retrieval pipelines using foundation-model representations and multiscale crops.

Deep Learning Research Assistant October 2022 – December 2022
Fraunhofer Institute for Integrated Circuits IIS Erlangen, Germany

- Developed and optimized deep learning pipelines in PyTorch for audio classification tasks, including speaker identification and acoustic event detection, achieving a 15% accuracy improvement.
- Designed robust preprocessing pipelines with noise reduction and spectrogram-based feature extraction, reducing training data errors by approximately 20%.

Software Developer May 2022 – September 2022
Semikron Danfoss Nuremberg, Germany

- Developed and maintained features and integrations for the company's Polarion ALM platform using Java, JavaScript, CSS, and HTML.

Software Developer Internship June 2018 – September 2018
Respina Networks and Company Beyond Tehran, Iran

- Developed and maintained web applications with JavaScript, React.js, jQuery, CSS, and HTML.

RESEARCH EXPERIENCE

Direct Visual Grounding by Directing Attention of Visual Tokens | WACV 2026

- Introduced an attention-based auxiliary loss to improve alignment between language tokens and image regions, guiding Vision–Language Models to focus on the correct image patches.

- Designed new benchmarks for geometric and grounding tasks, including line-intersection counting, path tracing, and pointing to object centers.
- Achieved an average **10.3%** accuracy improvement over baselines across tasks while enhancing interpretability of attention heatmaps, without requiring architectural changes or new labels.

Pathology Foundation Models for Whole-Slide Image Retrieval | npj Digital Medicine

- Evaluated whole-slide image retrieval across **9,387 TCGA diagnostic slides** spanning **17 organs and 60 diagnoses**.
- Benchmarked **10 retrieval pipelines**, comparing slide-level foundation models with supervised aggregation and patch-level retrieval approaches.
- Demonstrated that retrieval performance varies substantially across cancer types and that strong patch-level representations can perform comparably to more complex slide-level foundation models.

WHO-Aligned Retrieval for Fine-Grained Breast Pathology

- Developed an interpretable retrieval framework matching **149 private Mayo Clinic cases** to **405 WHO reference images** across **41 breast lesion categories**, evaluating retrieval on **31 shared categories**.
- Benchmarked **7 pathology foundation models** for fine-grained retrieval and classification.
- Found that using a single optimized crop scale outperformed the multiscale baseline at the top retrieval ranks, while incorporating classifier predictions further improved retrieval accuracy.

Logit Lens Supervision for Patch-Level Explanations in Vision-Language Models | eXCV Workshop at ECCV 2026

- Identified the loss of localized image information in visual tokens inside the language model and introduced Logit Lens Loss to preserve their connection to corresponding image regions.
- Used these visual representations to generate patch-level confidence maps, improving grounding by up to **11.4 cIoU points** while reducing hallucination and preserving VQA performance.

Angular Supervision for Visual Object Grounding in Autoregressive Vision-Language Models | WACV 2027 (Under Review)

- Identified distortions in explainability maps such as attention and Logit Lens caused by high-norm visual tokens, and introduced Angular Relevance to measure visual-token alignment using direction instead of magnitude.
- Introduced Angular Object-Aware Loss to preserve object information in visual tokens, improving segmentation by up to **9.4 cIoU points** and generalizing to unseen object categories.

PUBLICATIONS

Esmaeilkhani, P., Latecki, L. **Direct Visual Grounding by Directing Attention of Visual Tokens**. *IEEE/CVF Winter Conference on Applications of Computer Vision (WACV 2026)*.

Lei, T.†, **Esmaeilkhani, P.**†, …, Tizhoosh, H. R. **Multi-institutional Pan-Cancer Validation of Pathology Foundation Models for Whole-Slide Image Retrieval Using TCGA**. *npj Digital Medicine*, 2026. †Equal first authors.

Esmaeilkhani, P., Latecki, L. J. **Logit Lens Supervision for Patch-Level Explanations in Vision-Language Models**. *eXCV Workshop at ECCV 2026*.

Esmaeilkhani, P., Latecki, L. **Angular Supervision for Visual Object Grounding in Autoregressive Vision-Language Models**. Under Review (Second Round), *IEEE/CVF Winter Conference on Applications of Computer Vision (WACV 2027)*.

Russakoff, A., Miller, K., Mahzoon, V., **Esmaeilkhani, P.**, et al. **CourtsightTV: An Interactive Visualization Software for Labeling Key Basketball Moments**. *Proc. CIKM 2024*.

TECHNICAL SKILLS

Programming Languages: Python, Java, C++, MATLAB, SQL.

Frameworks & Toolkits: PyTorch, TensorFlow, DeepSpeed, Hugging Face.

Machine Learning: Computer Vision, Vision-Language Models, Natural Language Processing, Transformers, Foundation Models, Multimodal Learning, Self-Supervised Learning.

Tools & Development: NumPy, Pandas, Scikit-learn, Matplotlib, Git, Linux/Bash, Distributed Training.